

Wireless Data Transmission

The background image is a composite of a city skyline at night, likely New York City, with prominent skyscrapers like the Empire State Building. The scene is overlaid with a network of glowing, curved lines in shades of blue, purple, and pink, which represent wireless data transmission or signal paths. The lines originate from various points on the horizon and curve upwards and outwards, creating a sense of dynamic connectivity. The overall color palette is dominated by these vibrant, neon-like hues against the dark night sky and the city lights.

Alexa Hernandez

A large, metallic satellite dish antenna is shown from a low angle, looking up. The dish is composed of a complex grid of metal struts. It is mounted on a tall, lattice-like tower. The background is a clear, bright blue sky. The word "Background" is written in a large, white, sans-serif font across the middle of the image, partially obscuring the dish.

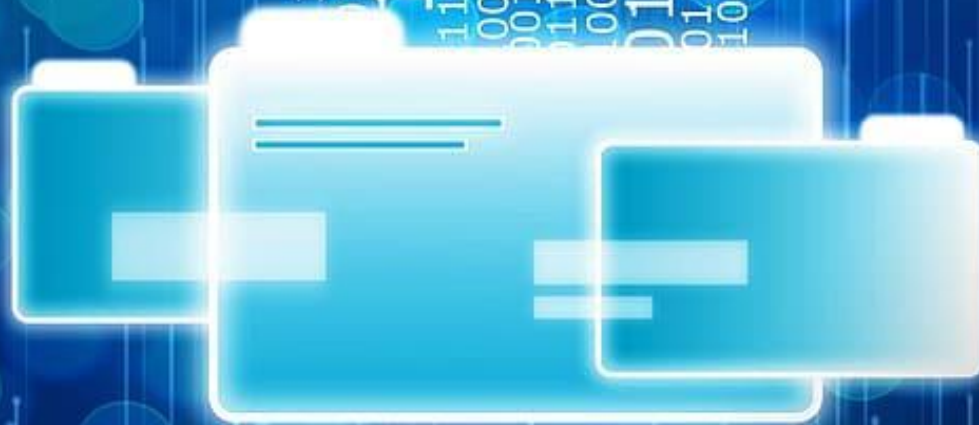
Background

Background

- “Wireless data communication is a technology that uses the radio spectrum to transmit signals through the atmosphere” (Gartner).
- The first concept of wireless data transmission was initially introduced through radio waves in 1896.
- Guglielmo Marconi invented the first telegraph in 1896.
- It was tested to send radio signals over the Atlantic Ocean.
- Of course, advancements have been made...



How does it work?



How it works

Procedure:

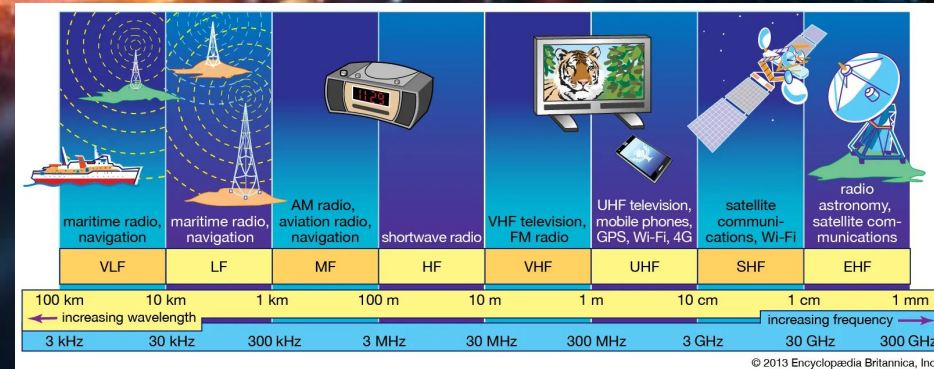
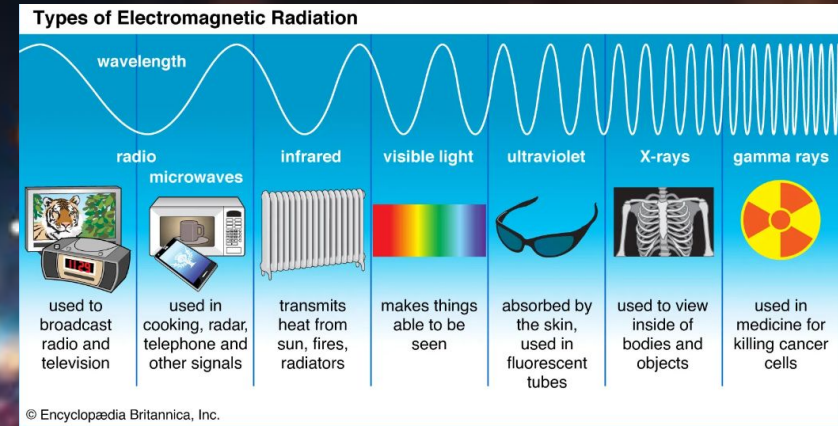
- Expanding on what we know from our class... Packet switching!
- Electromagnetic waves are used.

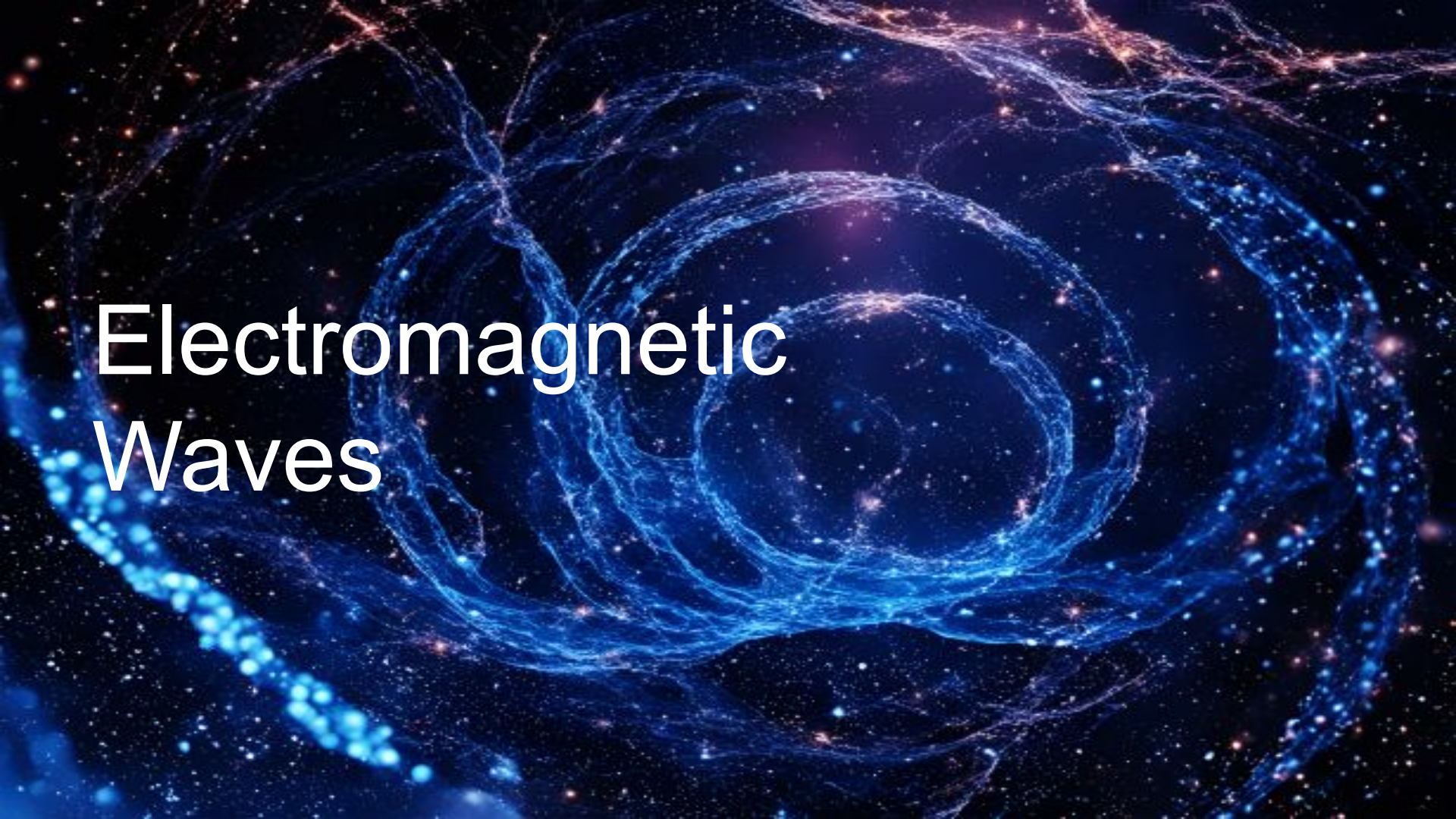
Abstraction/Imaging:

- Ripples of water
- Sound waves
- Light waves

Examples:

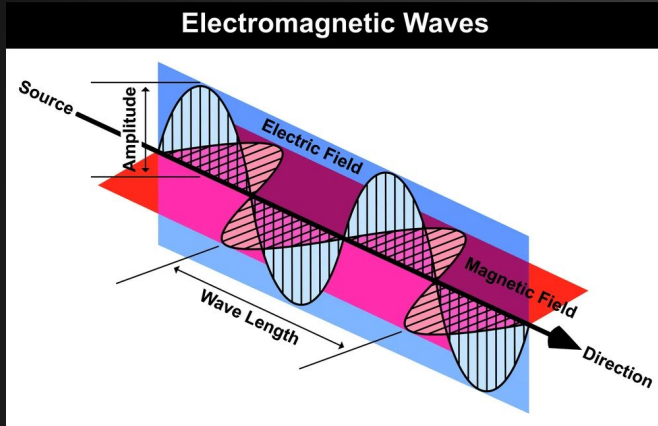
- 4G LTE, 5G, WiFi.





Electromagnetic Waves

Electromagnetic Waves



- Like radio waves, broken down data is encoded onto electromagnetic waves through protocols (TCP/IP).
- Amplitude and frequency is important!
- Modulating properties of waves (Shift Keying).
- Computers can interpret the modulated waves.
 - Ex. Start with number, Binary wave, Encoded Carrier wave, Binary wave, Decode, and back to number.

Sources

Images:

<https://www.borntoengineer.com/ucl-engineers-set-world-record-for-wireless-data-transmission-with-938-gbps-speeds>

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<https://www.britannica.com/science/electromagnetic-spectrum>

<https://www.britannica.com/topic/telecommunications-media/Radio-transmission>

<https://www.scienceabc.com/eyeopeners/do-all-photons-in-the-world-move-at-the-same-speed.html>

Information:

<https://www.gartner.com/en/information-technology/glossary/wireless-data-communication#:~:text=Wireless%20data%20communication%20is%20a,%2D%20or%20two%2Dway%20networks>.

<https://www.rfpage.com/history-of-wireless-communication-morse-code-to-5g-technology/#:~:text=The%20first%20wireless%20telegraph%20system, and%20holds%20800%20wireless%20patents>.

https://www.youtube.com/watch?v=WX_fSjgkqQ

Thank you!